



HITEC UNIVERSITY, TAXILA

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

Lab#7

Implementation of autocorrelation/cross correlation

Objective:

To design and implement Latches/Flipflops without using standard IC s, on Trainer board andLogisim.

Apparatus/Software:

- Python
- Anaconda

Theory:

Auto Correlation Function

Auto correlation function is a measure of similarity between a signal & its time delayed version. It is represented with R(k).

The auto correlation function of x(n) is given by

$$R_{11}(k)=R(k)=\sum_{n=-\infty}^{\infty} x_1(n)x_2(n-k)$$

Cross-correlation is a measure of similarity between two signals as a function of the time delay applied to one of the signals. It is commonly used in signal processing and communication systems to identify time delays and patterns in signals. In this section, we will discuss the formal theory and provide the Python code to compute the cross-correlation of two signals.

The cross-correlation between two discrete-time signals x(n) and y(n) is defined as:

$$R_{xy}(m) = \sum_n [x(n) * y(n+m)]$$

where m is the time shift, and R_{xy}(m) is the cross-correlation of x and y at time shift m.

The cross-correlation is a measure of the similarity between the two signals as a function of the time shift. If the two signals are similar, the cross-correlation will have high values at small time shifts. If the two signals are dissimilar, the cross-correlation will have low values at all time shifts.

Algorithm:

Step I : Give input sequence x[n].

Step II : Give impulse response sequence h (n)

Step III: Find auto correlation using the matlab command xcorr. Step IV: Plot x[n], h[n], z[n].

Step V: Display the results

Code



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```
import numpy as np
import matplotlib.pyplot as plt

# Two input sequences
x = np.array([1, 2, 3, 4, 5])

# Plot input sequence
plt.subplot(1, 2, 1)
plt.stem(x)
plt.xlabel('n')
plt.ylabel('x(n)')
plt.title('input sequence')

# Auto-correlation of input sequence

z = np.correlate(x, x, mode='full')
print('The values of z are = ')
print(z)

# Plot auto-correlation of input sequence
plt.subplot(1, 2, 2)
plt.stem(z)
plt.xlabel('n')
plt.ylabel('z(n)')
plt.title('auto correlation of input sequence')

plt.show()

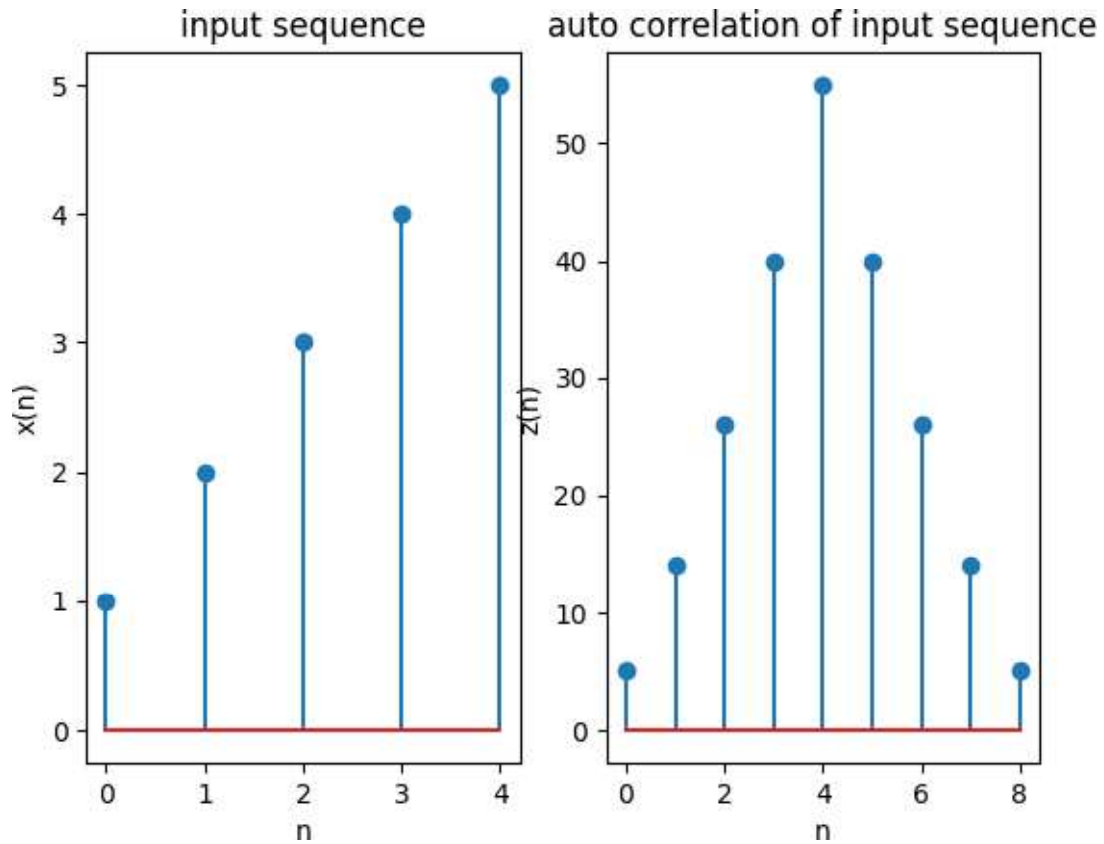
output
```



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```
import numpy as np
import matplotlib.pyplot as plt

# Two input sequences
x = np.array([1, 2, 3, 4, 5])
y = np.array([3, 4, 5, 6, 7])

# Compute cross-correlation
r_xy = np.correlate(x, y, mode='full')

# Plot signals and cross-correlation
plt.subplot(3, 1, 1)
plt.stem(x)
plt.xlabel('Sample index')
plt.ylabel('Amplitude')
plt.title('Signal x')

plt.subplot(3, 1, 2)
plt.stem(y)
plt.xlabel('Sample index')
plt.ylabel('Amplitude')
plt.title('Signal y')

plt.subplot(3, 1, 3)
plt.stem(r_xy)
plt.xlabel('Lag')
plt.ylabel('Correlation')
plt.title('Cross-correlation of x and y')

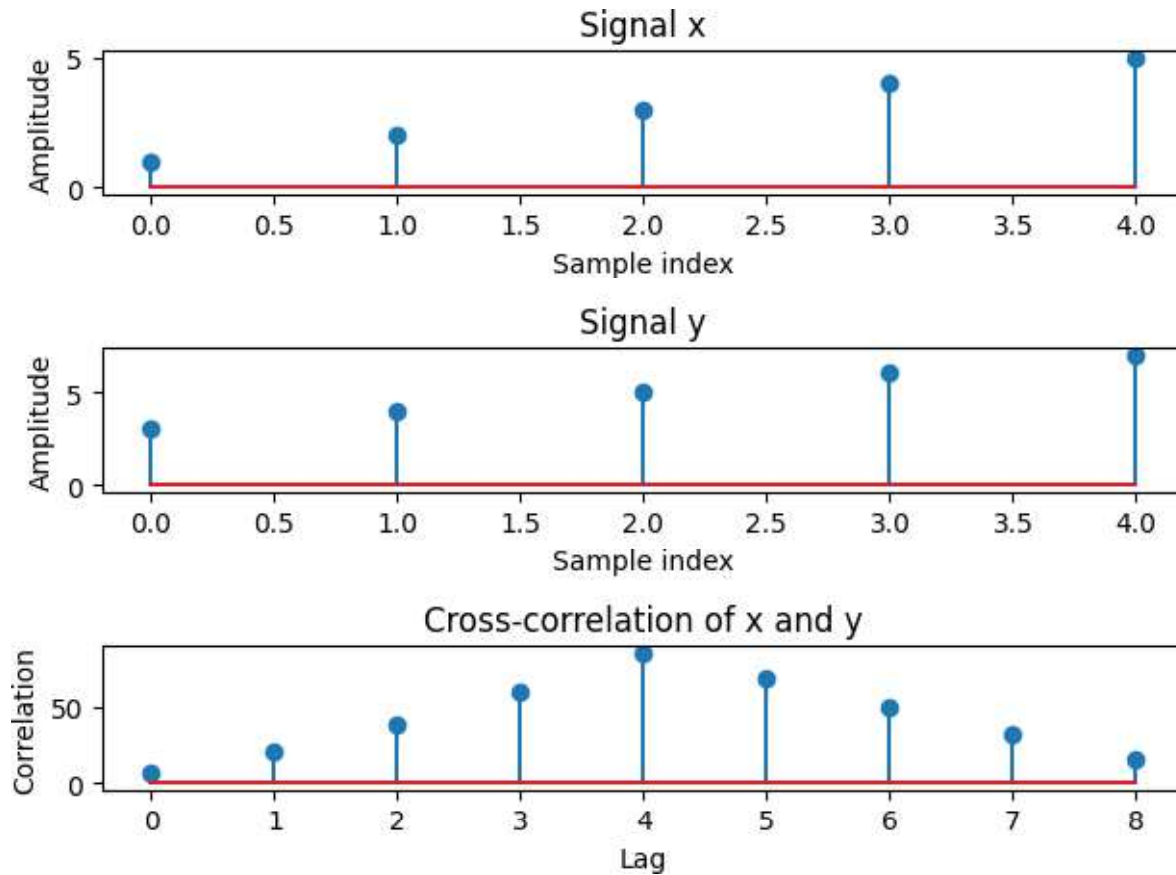
plt.tight_layout()
plt.show()
```



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Lab Report Evaluation: (To be submitted in printed form next week before lab)

➤ Lab Task

Find the cross correlation of the sequences

$$x(n) = \begin{Bmatrix} 3, 5, 1, 2 \\ \uparrow \end{Bmatrix} \text{ and } h(n) = \begin{Bmatrix} 1, 4, 3 \\ \uparrow \end{Bmatrix}$$

Find the cross correlation of two finite length sequences:

$$x(n) = \{2, 3, 1, 4\} \text{ and } y(n) = \{1, 3, 2, 1\}$$

Find the periodic autocorrelation of the sequences

$$(a) x(n) = \{1, 3, 0, 4\} \quad (b) y(n) = \{2, 1, -2, 1\}$$



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Find the discrete periodic cross correlation $R_{xyp}(n)$ and $R_{yxp}(n)$ of the whose first period is given by

$$x(n) = \left\{ \underset{\uparrow}{1}, 3, 0, 4 \right\} \quad \text{and} \quad y(n) = \left\{ \underset{\uparrow}{2}, 1, -2, 1 \right\}$$

Important note: Lab Report must be according to the Lab Report Format available on LMS and must be submitted on time. Two similar reports will get zero marks. So, don't try to copy your fellow reports. Lab report must be submitted in group of three maximum